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# Rate of the weak reaction $s+u \rightarrow u+d$ in quark matter

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The weak reaction $s+u \rightarrow u+d$ and its inverse play important roles in the attempts to detect strangelets from relativistic heavy-ion collisions, in the burning of neutron stars into strange stars, in the formation and evaporation of quark nuggets in the early Universe, and, through the bulk viscosity, in determining the stability properties and maximum rotation velocity of neutron stars and/or strange stars. Exact expressions for these rates are given in the case of zero temperature and strange quark mass, and numerical calculations and fits to analytic approximations are given for the general case. Pauli blocking and finite-temperature effects are crucial. There is no "typical" time scale for the reactions.

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## I. INTRODUCTION

Formation of stable or metastable strangelets composed of comparable numbers of up, down, and strange quarks has been proposed as a possible signature of the formation of a quark-gluon plasma in relativistic heavy-ion collisions [1]. The experimental signal is crucially dependent on the rate of weak reactions among the participating quarks—in particular on the reaction

$$s+u \rightarrow u+d \tag{1a}$$

and its inverse reaction

$$u+d \rightarrow s+u \ . \tag{1b}$$

These very same reactions determine the speed with which a burning front can propagate in a neutron star in the transformation of a neutron star into a neutron star with a quark core, or perhaps even a strange star, given that strange quark matter is absolutely stable (otherwise the high internal pressure can favor transformation of part of the star) [1]. The fact that the weak reactions in Eqs. (1) have rates comparable to the characteristic rotation and pulsation rates of neutron stars means that these reactions can be decisive for determination of the bulk viscosity of these stars, and this bulk viscosity again may determine the stability properties and maximum rotation rates of the stars [1]. Finally, Eq. (1a) limits the rate at which cosmologically produced nuggets of strange quark matter may evaporate in the early Universe [1].

All in all there is a general interest in calculating the rate of reactions (1a) and (1b), including the decisive effects of Pauli blocking and finite temperature. Such an attempt was done previously in Ref. [2], but the matrix element in that paper was wrong. Recently several authors have found approximate expressions in certain limits (typically; low temperature, zero strange quark mass, and almost-equal quark chemical potentials).

In the present investigation I shall present a general recipe for the rate calculation, give an exact analytical expression for arbitrary chemical potentials, but zero mass and temperature, and give numerical results for different parameter sets, compared with the analytical approximations available. The calculations are for the weak interaction part only. Strong corrections may play a role, but there seems to be no way at present of estimating the size.

## II. GENERAL EXPRESSION FOR THE RATE

The quarks involved in Eqs. (1) are described by Fermi-Dirac distributions of the form

$$f_q(E_q)=\{1+\exp[(E_q-\mu_q)/T]\}^{-1}, \quad q=u,d,s \ , \tag{2}$$

where $\mu_q$ are the chemical potentials (usually of order a few hundred MeV in the systems of interest). Unless otherwise noted, all equations assume $\hbar=c=k_B=1$. Equation (2) corresponds to a number density in phase space of

$$n_q(E_q)=6f_q(E_q)/(2\pi)^3 \ , \tag{3}$$

where the factor 6 stems from 2 spins and 3 colors.

As argued in [2] the rate per volume of reaction (1a) is given by

$$\Gamma(su_1 \rightarrow u_2 d)=\frac{1}{2}\int 36(2\pi)^{-5}\frac{1}{2E_s}\frac{1}{2E_1}\frac{1}{2E_2}\frac{1}{2E_d}\delta^4(P_s+P_1-P_2-P_d)$$

$$\times|M(su_1 \rightarrow u_2 d)|^2 f_1(E_1)f_s(E_s)[1-f_d(E_d)][1-f_2(E_2)]d^3p_1 d^3p_s d^3p_d d^3p_2 \tag{4}$$

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with the matrix element summed over final spins and averaged over initial spins

$$|M(su_1\to u_2d)|^2=64G_F^2\sin^2\theta_C\cos^2\theta_C(P_1\cdot P_s)(P_2\cdot P_d) \tag{5}$$

(another combination of four-momenta was erroneously used in [2]). Here $P_q=(E_q,\mathbf{p}_q)$ is the four-momentum of quark $q$, and $G_F$ is the Fermi constant. $\theta_C$ is the Cabibbo angle. A factor of $\frac{1}{2}$ is explicitly shown in Eq. (4) to take into account that only left-handed helicity states of the $u_1$ quarks couple to the $W^-$, which mediates the transformation. Henceforth we neglect the masses of up and down quarks, so that $E_1=p_1$, $E_2=p_2$, $E_d=p_d$, $E_s=(p_s^2+m_s^2)^{1/2}$.

The rate per $s$ quark is obtained from Eq. (4) by dividing by the number density of strange quarks,

$$N_s=\int n_s\,d^3p_s\ . \tag{6}$$

The rate per volume of reaction (1b) is found by interchanges of indices. It can easily be shown from the Fermi factors that

$$\Gamma(u_2d\to su_1)=\exp\left[\frac{\mu_d-\mu_s}{T}\right]\Gamma(su_1\to u_2d)\ , \tag{7}$$

so that the net rate of transforming $s$ quarks into $d$ quarks is

$$\Gamma(s\to d)=\left[1-\exp\left[\frac{\mu_d-\mu_s}{T}\right]\right]\Gamma(su_1\to u_2d)\ . \tag{8}$$

The four-momentum products give

$$(P_1\cdot P_s)(P_2\cdot P_d)=E_dE_sE_1E_2\left[1-\frac{p_s}{E_s}\cos\theta_{1s}\right](1-\cos\theta_{2d})\ , \tag{9}$$

where $\theta_{ij}$ denotes the angle between quarks $i$ and $j$.

The four-dimensional $\delta$ function can be written as

$$\delta^4(P_s+P_1-P_2-P_d)=\delta(E_s+E_1-E_2-E_d)\times\delta^3(\mathbf{p}_s+\mathbf{p}_1-\mathbf{p}_2-\mathbf{p}_d)\ , \tag{10}$$

with the spherical-coordinate representation

$$\delta^3(\mathbf{p}_s+\mathbf{p}_1-\mathbf{p}_2-\mathbf{p}_d)=\frac{1}{|\mathbf{p}_s+\mathbf{p}_1|^2}\delta(|\mathbf{p}_s+\mathbf{p}_1|-|\mathbf{p}_2+\mathbf{p}_d|)\delta(\phi-\phi')\delta(\cos\theta_{2d}-g(\theta_{1s}))\ , \tag{11}$$

$$g(\theta_{1s})=\frac{p_1^2+p_s^2-p_2^2-p_d^2+2p_1p_s\cos\theta_{1s}}{2p_2p_d}\ . \tag{12}$$

Using Eqs. (9)–(12) in Eq. (4), one gets (with a factor $2\pi$ from the $\phi$ integration)

$$\Gamma(su_1\to u_2d)=\frac{72}{(2\pi)^5}G_F^2\sin^2\theta_C\cos^2\theta_C\times\int_0^\infty dE_1\int_0^\infty dE_2\int_{m_s}^\infty dE_s\int_0^\infty dE_d\,\delta(E_1+E_s-E_2-E_d)\times f_1(E_1)f_s(E_s)[1-f_d(E_d)][1-f_2(E_2)]\frac{E_1}{E_s}[h(1)-h(x_{\min})] \tag{13}$$

with

$$x_{\min}=\min\left\{1,\max\left\{-1,\frac{m_s^2+2E_1E_s-4E_2E_d}{2E_1p_s}\right\}\right\}\ , \tag{14}$$

$$h(x)=\left[\frac{b(3g^2x^2-4fgx+8f^2)}{15g^2}-\frac{c(gx-2f)}{3g}+d\right](f+gx)^{1/2}\ , \tag{15}$$

$$\begin{aligned}b&=2E_1E_sp_s^2\ ,\\ c&=m_s^2p_sE_s+4E_1p_sE_s^2\ ,\\ d&=m_s^2E_s^2+2E_1E_s^3\ ,\\ f&=E_1^2+p_s^2\ ,\\ g&=2E_1p_s\ .\end{aligned} \tag{16}$$

Finally, after integrating over $dE_2$, one gets the expression

$$\Gamma(su_1\to u_2d)=\frac{72}{(2\pi)^5}G_F^2\sin^2\theta_C\cos^2\theta_C \times\int_{m_s}^{\infty}dE_s\left[\int_0^{p_s}dE_1\left[\int_{(E_s-p_s)/2}^{(2E_1+E_s-p_s)/2}dE_d\,\mathcal{C}(1,y)+\int_{(2E_1+E_s-p_s)/2}^{(E_s+p_s)/2}dE_d\,\mathcal{C}(1,-1)+\int_{(E_s+p_s)/2}^{(2E_1+E_s+p_s)/2}dE_d\,\mathcal{C}(1,y)\right]+\int_{p_s}^{\infty}dE_1\left[\int_{(E_s-p_s)/2}^{(E_s+p_s)/2}dE_d\,\mathcal{C}(1,y)+\int_{(E_s+p_s)/2}^{(2E_1+E_s-p_s)/2}dE_d\,\mathcal{C}(1,-1)+\int_{(2E_1+E_s-p_s)/2}^{(2E_1+E_s+p_s)/2}dE_d\,\mathcal{C}(1,y)\right]\right] \tag{17}$$

with

$$\mathcal{C}(1,y)=f_1(E_1)f_s(E_s)[1-f_d(E_d)][1-f_2(E_1+E_s-E_d)][h(1)-h(y)]E_1/E_s\ , \tag{18}$$

$$y=\frac{m_s^2+2E_1E_s-4E_d(E_1+E_s-E_d)}{2E_1p_s}\ . \tag{19}$$

## III. THE LIMIT $T=m_s=0$

The evaluation of Eq. (17) simplifies tremendously at zero temperature, because the Fermi-Dirac distributions become step functions, and $f_1f_s(1-f_d)(1-f_2)=1$ when $\mu_d<E_d<E_1+E_s-\mu_u$, $\max(0,\mu_u+\mu_d-E_s)<E_1<\mu_u$, $\mu_d<E_s<\mu_s$. Otherwise the rate is zero.

Neglecting the strange quark mass (i.e., $m_s=0$) reduces the problem further. Introducing $z\equiv\mu_d/\mu_u$ and $v\equiv(\mu_s-\mu_d)/\mu_u$ one finds (for $\mu_u<\mu_d$, $\mu_s<\mu_u+\mu_d$)

$$\Gamma(su_1\to u_2d)=\frac{72}{(2\pi)^5}G_F^2\sin^2\theta_C\cos^2\theta_C\int_{\mu_d}^{\mu_s}dE_s\int_{\mu_u+\mu_d-E_s}^{\mu_u}dE_1\int_{\mu_d}^{E_1+E_s-\mu_u}dE_d\left(\tfrac{16}{3}E_1^3E_s^2+\tfrac{8}{15}E_1^5+\tfrac{8}{3}E_1^4E_s\right)$$
$$=\frac{16}{5\pi^5}G_F^2\sin^2\theta_C\cos^2\theta_C\mu_u^5(\mu_s-\mu_d)^3 \times\left\{\tfrac{1}{16}[1+5z+10z^2]+\tfrac{5}{32}v[1+4z-3z^2]-\tfrac{3}{16}v^2[-1+3z-z^2] +\tfrac{1}{32}v^3[-6+8z-z^2]+\tfrac{5}{112}v^4[2-z]-\tfrac{15}{896}v^5\right\} \tag{20}$$

and (for $\mu_u<\mu_d$, $\mu_s>\mu_u+\mu_d$)

$$\Gamma(su_1\to u_2d)=\frac{72}{(2\pi)^5}G_F^2\sin^2\theta_C\cos^2\theta_c\left[\int_{\mu_d}^{\mu_u+\mu_d}dE_s\int_{\mu_u+\mu_d-E_s}^{\mu_u}dE_1\int_{\mu_d}^{E_1+E_s-\mu_u}dE_d\left(\tfrac{16}{3}E_1^3E_s^2+\tfrac{8}{15}E_1^5+\tfrac{8}{3}E_1^4E_s\right)+\int_{\mu_u+\mu_d}^{\mu_s}dE_s\int_0^{\mu_u}dE_1\int_{\mu_d}^{E_1+E_s-\mu_u}dE_d\left(\tfrac{16}{3}E_1^3E_s^2+\tfrac{8}{15}E_1^5+\tfrac{8}{3}E_1^4E_s\right)\right]$$
$$=\frac{16}{5\pi^5}G_F^2\sin^2\theta_C\cos^2\theta_c\,\mu_u^5(\mu_s-\mu_d)^3\left\{\tfrac{15}{64}v+\tfrac{1}{16}[1+10z]+\tfrac{15}{32}z^2v^{-1} -\tfrac{1}{112}[1+7z+21z^2]v^{-2}+\tfrac{1}{896}[3+16z+28z^2]v^{-3}\right\}\ . \tag{21}$$

In the limit where $\mu_d=\mu_u$ and $v\to0$ the rate becomes

$$\Gamma(su_1\to u_2d)\approx\frac{16}{5\pi^5}G_F^2\sin^2\theta_C\cos^2\theta_C\mu_u^5(\mu_s-\mu_d)^3\ , \tag{22}$$

an expression in agreement with earlier work in [3]–[5], but a factor 24 larger than the value quoted in [6].

Figure 1 shows the rate from Eqs. (20) and (21) as a function of $z$ and $v$ normalized to the approximative expression in Eq. (22). The actual rate equals the approximate expression for $z=1$, $v=0$. It deviates most for high $z$ and $v\to0$, but rarely by a large factor when $T=m_s=0$.

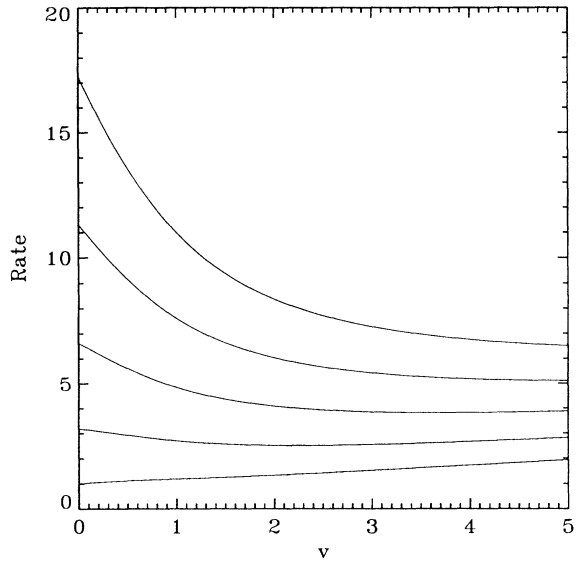



FIG. 1. $\Gamma(su_1 \to u_2 d)$ at $T = m_s = 0$ from Eqs. (20) and (21) normalized by the approximation in Eq. (22) for $z = 1, 2, 3, 4$, and 5 (lower to upper curve).

## IV. THE LIMIT OF SMALL $T$

The net rate for conversion of strange into down quarks has been expanded for $m_s = 0$, low temperature, $\mu_d = \mu_u, \mu_s \to \mu_d$, in Refs. [3] and [5], with the result

$$\Gamma(s \to d) \approx \frac{16}{5\pi^5} G_F^2 \sin^2\theta_C \cos^2\theta_C \times \mu_u^5 (\mu_s - \mu_d)[(\mu_s - \mu_d)^2 + 4\pi^2 T^2] \,. \quad (23)$$

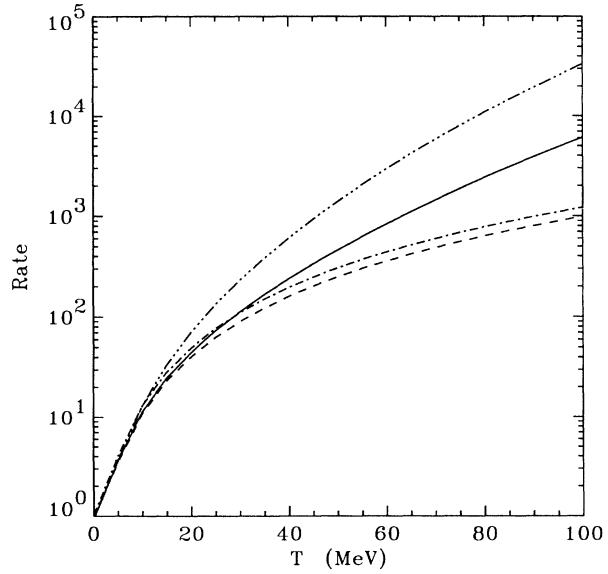



FIG. 2. $\Gamma(su_1 \to u_2 d)$ at $m_s = 0$ normalized by Eq. (22), for $\mu_u = \mu_d = 200$ MeV, $\mu_s = 220$ MeV. The curves are (from top to bottom): The numerical rate from Eq. (17); the net rate from Eq. (8); the approximation combining Eqs. (20), (23), and (24); the approximation using Eq. (23) only.

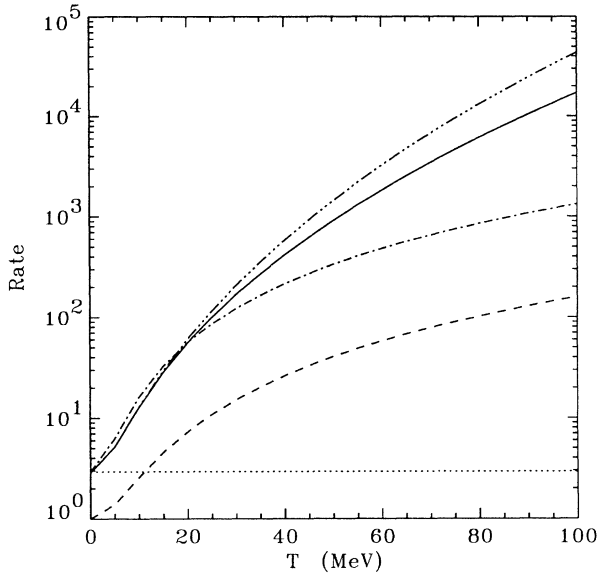



FIG. 3. $\Gamma(su_1 \to u_2 d)$ at $m_s = 0$ normalized by Eq. (22), for $\mu_u = 100$ MeV, $\mu_d = 200$ MeV, $\mu_s = 250$ MeV. The horizontal line is the $T = 0$ approximation from Eq. (20). The other curves are as in Fig. 2.

As shown in Fig. 2 this approximation works well for temperatures small compared to $\mu_s - \mu_d$, as long as $\mu_u = \mu_d$. However, the deviation becomes notable when $\mu_u \neq \mu_d$—from a factor of a few at $T \to 0$ to a factor of 100 at $T = 100$ MeV in the case shown in Fig. 3.

Sawyer [7] seems to lack an overall factor of 3 in his calculation of $\Gamma(s \to d)$. Furthermore he expands only to first order in $(\mu_s - \mu_d)$ and therefore misses the temperature-independent term, which actually dominates for $2\pi T < (\mu_s - \mu_d)$. The consequences of this for the bulk viscosities calculated in [7] are treated elsewhere [8].

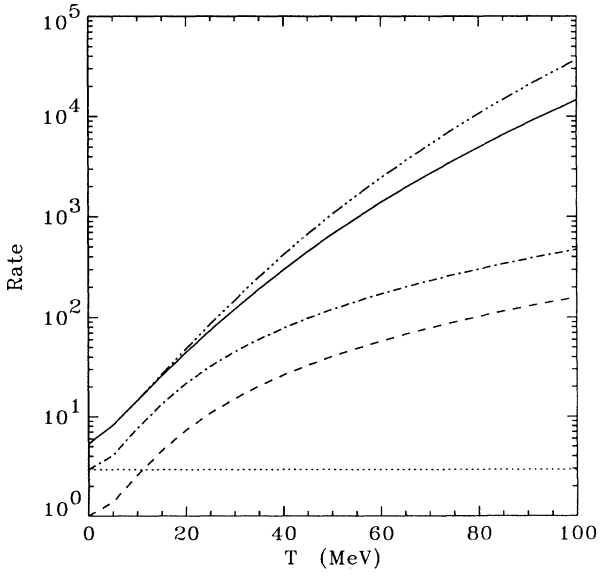



FIG. 4. Same as Fig. 3, except for $m_s = 200$ MeV.

Reference [7] gives an explicit expression for the $T^2$ term which includes the corrections due to different chemical potentials and finite $m_s$. Effectively the $T^2$ term in Eq. (23) should be multiplied by [9]

$$[-\tfrac{1}{128}v^5-\tfrac{5}{64}zv^4+\tfrac{5}{64}(2-z^2)v^3 +\tfrac{5}{16}(1+3z)v^2+\tfrac{5}{16}(4z+3z^2)v+\tfrac{5}{4}z^2-\tfrac{1}{4}] , \quad (24)$$

where $m_s=0$, and by

$$\tfrac{5}{256}(\tfrac{3}{5}(32+y^5)-(8+y^3)\{4+[(\mu_s+\mu_d)/\mu_u]^2-m_s^2/\mu_u^2\} +(24+12y)\{[(\mu_s+\mu_d)/\mu_u]^2-m_s^2/\mu_u^2\}) , \quad (25)$$

where $y=[(\mu_s^2-m_s^2)^{1/2}-\mu_d]/\mu_u$, when $m_s\neq 0$.

Figures 3 and 4 illustrate how the fits are significantly improved by inclusion of this correction when added to the full zero-temperature terms from Eqs. (20) and (21). At nonzero $m_s$ this correction is not fully adequate, most likely due to the assumption of $m_s=0$ in the zero-temperature terms.

## V. CONCLUSION

Numerical results for the weak reaction $us \rightarrow du$ and its inverse have been presented and compared to existing and new analytical limits. The net rate for strange to down quark conversion in quark matter is shown to depend strongly on Pauli blocking and finite-temperature effects. The latter are decisive for $T > |\mu_s-\mu_d|/2\pi$, whereas Eq. (22) can be applied for lower temperatures when $m_s$ is small and the chemical potentials of comparable magnitude. Equations (20) and (21) are exact when $T=0, m_s=0$.

Figures 5 and 6 give examples of rates per $s$ quark per second. Figure 5 shows the (not terribly strong) dependence on $m_s$, and Fig. 6 illustrates among other things that the rate for $su \rightarrow ud$ is an increasing function of temperature at fixed chemical potentials, and an increasing function of $\mu_s$ at fixed temperature. The two figures also illustrate that there is no such thing as a "typical" time scale for the reaction. The rate of course goes to zero at low temperatures when $\mu_s \rightarrow \mu_d$, whereas high temperatures lift the degeneracy, leading to reduced Pauli blocking and higher rates. Thus the time scale may be anything from zero to infinity, with the range $10^{-10}$–$10^{-5}$ s covered by the examples shown in Figs. 5 and 6.

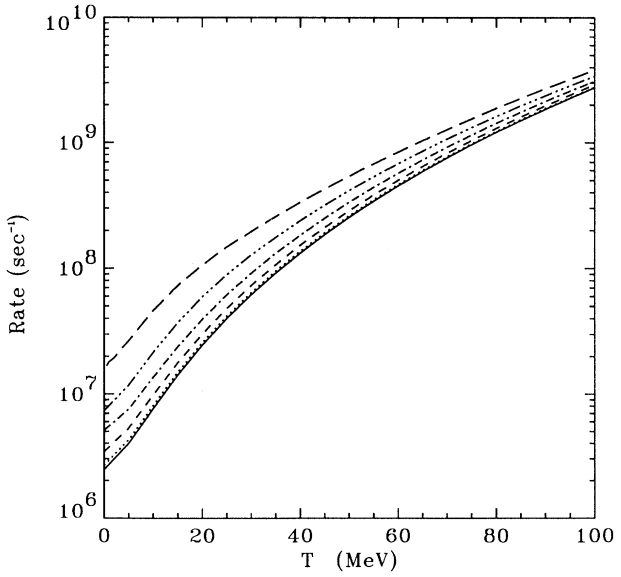



FIG. 5. Rate per second per $s$ quark of the process $su_1 \rightarrow u_2 d$ for $\mu_u=\mu_d=200$ MeV, $\mu_s=250$ MeV for $m_s$ going from 0 to 250 MeV in steps of 50 MeV (lower to upper curve).

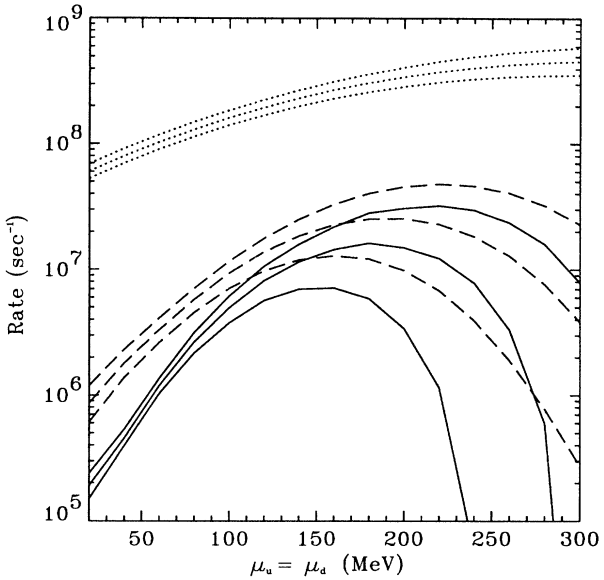



FIG. 6. Rate per second per $s$ quark of the process $su_1 \rightarrow u_2 d$ as a function of $\mu_u=\mu_d$. Dotted curves are at $T=50$ MeV, dashed curves at $T=10$ MeV, and solid curves at $T=0$. For each temperature the curves (upper to lower) correspond to $\mu_s=350$, 300, and 250 MeV, respectively. $m_s=100$ MeV.

The rates derived are important for several applications in heavy-ion collision physics and astrophysics. Among the consequences is a revised bulk viscosity of strange quark matter [8]. In the context of evaporation of cosmologically produced quark nuggets, the rates derived above are slightly higher than those used in [2] and [10], but the conclusions of those papers are unchanged. In relation to strangelets produced in heavy-ion collisions, the weak reactions discussed here may be crucial for the fate of the strangelets in their travel from the place of production to the detector. The rates have to be included self-consistently in studies hereof.

Readers interested in getting rates for specific parameter values may contact the author via electronic mail: jesm@dfi.aau.dk.

## ACKNOWLEDGMENTS

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[1] For reviews and references, see *Strange Quark Matter in Physics and Astrophysics,* Proceedings of the International Workshop, Aarhus, Denmark, 1991, edited by J. Madsen and P. Haensel [Nucl. Phys. B (Proc. Suppl.) **24B** (1991)].
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